

Histogram:

A histogram is a graphical representation of the distribution of a numerical data. It is an estimate of the probability distribution of a continuous variable and used to visualize the shape, central tendency, and variability of a dataset.

Agus :- {11, 12, 14, 18, 24, 26, 30, 35, 36, 37, 40, 41,

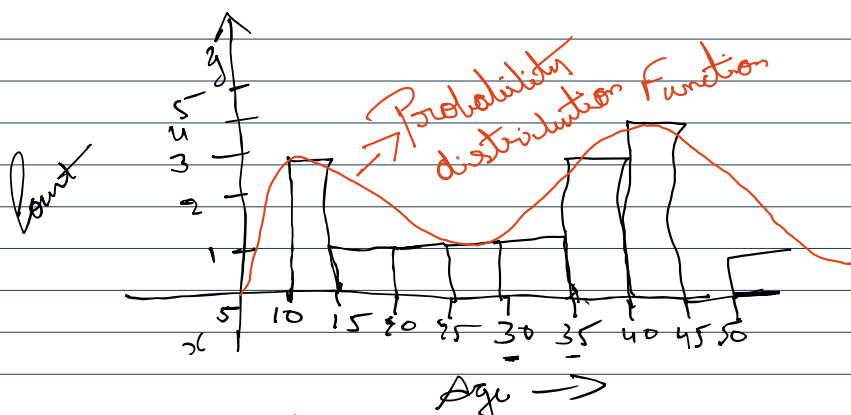
42, 43, 50}

11 Histogram

Bins $\Rightarrow 50 \Rightarrow 10$

$\frac{50}{10} = 5 \Rightarrow$ Bin Size

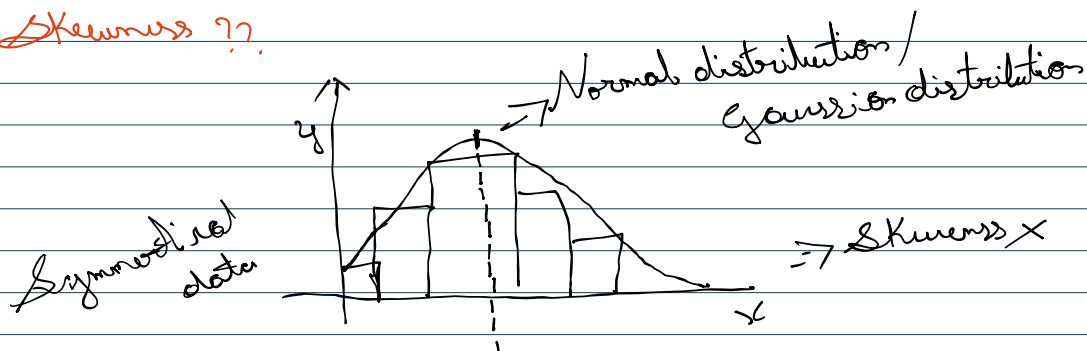
0-5 $\rightarrow 0$
 5-10 $\rightarrow 0$
 10-15 $\rightarrow 3$
 15-20 $\rightarrow 1$
 20-25 $\rightarrow 1$
 25-30 $\rightarrow 1$
 30-35 $\rightarrow 2$
 35-40 $\rightarrow 3$
 40-45 $\rightarrow 4$
 45-50 $\rightarrow 0$



Kernel density estimator

30 35
 30 35
 Agus :- {11, 12, 14, 18, 24, 26, 30, 35, 36, 37, 40, 41, 42, 43, 50}

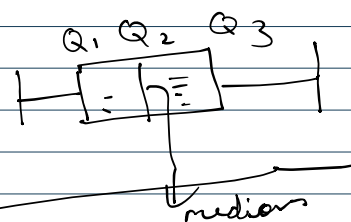
Skewness ??



Normal dist

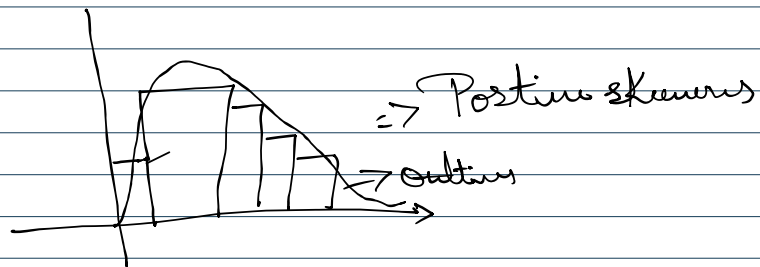


Mean = Median = Mode

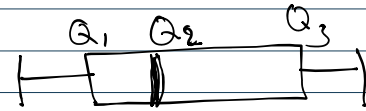


$Q_3 - Q_2 \approx Q_2 - Q_1$

② Right Skewed

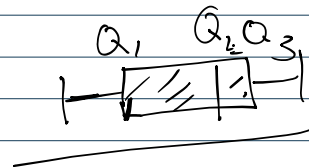
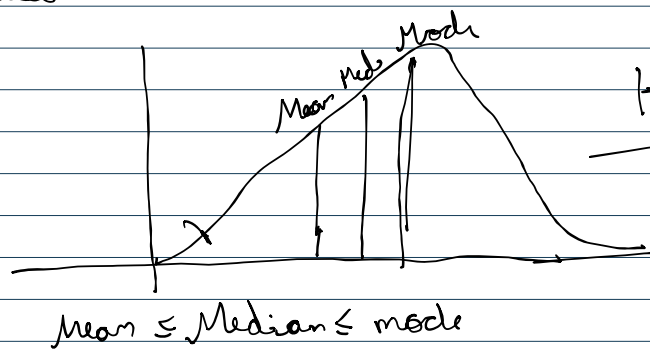


$$\text{Mean} > \text{Median} > \text{Mode}$$



$$Q_3 - Q_2 > Q_2 - Q_1$$

2) Left skewed



$$Q_2 - Q_1 > Q_3 - Q_2$$